

Price Discrimination and Monopoly: Linear Pricing

Introduction

- **Prescription drugs are cheaper in Canada than the United States**
- **Textbooks are generally cheaper in Britain than the United States**
- **Examples of *price discrimination***
 - **presumably profitable**
 - **should affect market efficiency: not necessarily adversely**
 - **is price discrimination necessarily bad – even if not seen as “fair”?**

Feasibility of price discrimination

- **Two problems confront a firm wishing to price discriminate**
 - *identification*: the firm is able to identify demands of different types of consumer or in separate markets
 - *easier in some markets than others: e.g tax consultants, doctors*
 - *arbitrage*: prevent consumers who are charged a low price from reselling to consumers who are charged a high price
 - *prevent re-importation of prescription drugs to the United States*
- **The firm then must choose the type of price discrimination**
 - first-degree or personalized pricing
 - second-degree or menu pricing
 - third-degree or group pricing

Third-degree price discrimination

- **Consumers differ by some observable characteristic(s)**
- **A uniform price is charged to all consumers in a particular group – linear price**
- **Different uniform prices are charged to different groups**
 - “kids are free”
 - subscriptions to professional journals e.g. *American Economic Review*
 - airlines
 - *the number of different economy fares charged can be very large indeed!*
 - early-bird specials; first-runs of movies

Third-degree price discrimination 2

- The pricing rule is very simple:
 - consumers with low elasticity of demand should be charged a high price
 - consumers with high elasticity of demand should be charged a low price
- Here we assume that the hard work is already done:
 - Data scientist(s) have already estimated properly the demand function using 2SLS, DID and/or RCT

Quantity optimization

Let us first introduce a simple profit optimization model...

Firm's profit:

$$\pi(Q) = P(Q)Q - F - CQ$$
$$\Rightarrow MR - MC = 0 \Rightarrow MR = MC$$

where:

- *Quantity demanded: Q*
- *Price (depends on Q): $P(Q)$*
- *Marginal cost: C*
- *Fixed cost: F*
- *Revenue: $R = P(Q)Q$*
- *Total cost: $TC(Q) = F + CQ$*

What could be wrong in the profit equation? So, what are the model risks here?

Quantity optimization, Cont'd

Assuming $\pi(Q)$ is concave then the first order condition (FOC):

$$\begin{aligned}\pi'(Q) &= P'(Q)Q + P(Q) - C = 0 \\ \Rightarrow \text{MR} - \text{MC} &= 0 \Rightarrow \text{MR} = \text{MC}\end{aligned}$$

where:

- $\pi'(Q) = d\pi(Q)/dQ$
- *Marginal revenue: $MR = dR/dQ$*
- *Marginal cost: $MC = dTC/dQ$*

*So, optimal quantity Q is at the point, where $MR = MC$ (if $\pi(Q) > 0$).
This is your main pricing condition in this class!*

Third degree price discrimination: example

- Amusement park tickets for children vs adults
- Demand:
 - Adults: $P_A = 36 - 4Q_A$
 - Children: $P_C = 24 - 4Q_C$
- Marginal cost constant in each market
 - $MC = \$4$
- What are the model risks here?
- Reminder on the hard work being done: The data scientist(s) have already estimated the demand function using 2SLS, DID, and/or RCT.

The example: no price discrimination (NPD)

- **Suppose that the same price is charged in both markets**
- **Use the following procedure:**
 - **calculate aggregate demand in the two markets**
 - **identify marginal revenue for that aggregate demand**
 - **equate marginal revenue with marginal cost to identify the profit maximizing quantity**
 - **identify the market clearing price from the aggregate demand**
 - **calculate demands in the individual markets from the individual market demand curves and the equilibrium price**
- **Let's continue with our example, easier to see...**

The example (NPD, Cont'd)

Adults: $P_A = 36 - 4Q_A$

Invert this:

$$Q_A = 9 - P/4 \text{ for } P \leq \$36$$

Children: $P_C = 24 - 4Q_C$

Invert:

$$Q_C = 6 - P/4 \text{ for } P \leq \$24$$

Aggregate these demands

$$Q = Q_A + Q_C = 9 - P/4 \text{ for } \$24 \leq P \leq \$36$$

$$Q = Q_A + Q_C = 15 - P/2 \text{ for } P < \$24$$

At these prices
only the adult
market is active

Now both
markets are
active

The example (NPD, Cont'd)

Invert the direct demands

$$P = 36 - 4Q \text{ for } Q \leq 3$$

$$P = 30 - 2Q \text{ for } Q > 3$$

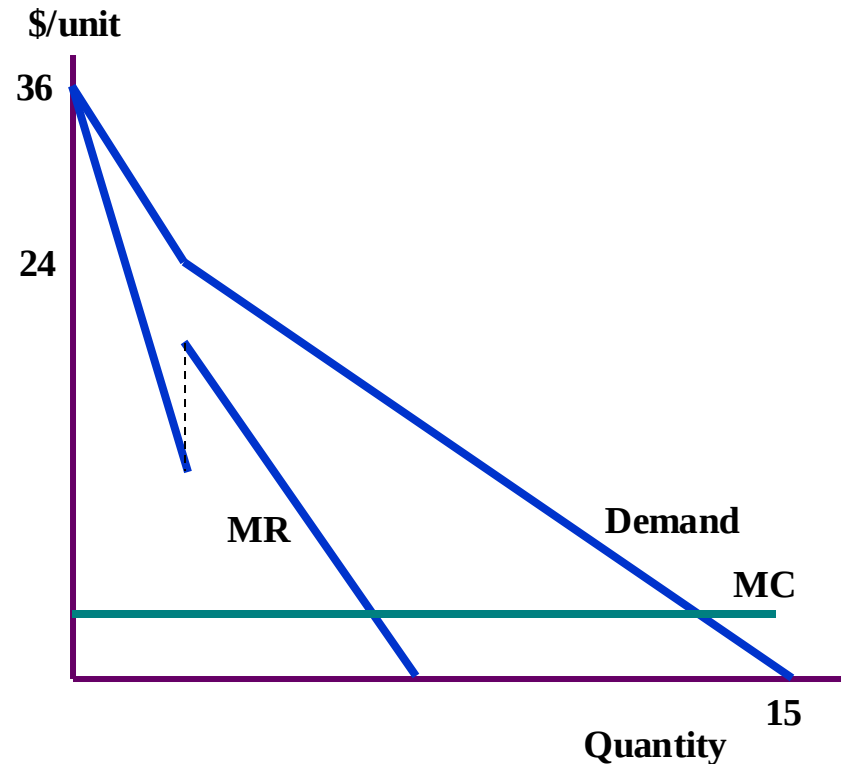
What happens if $Q > 3$?

Or if $Q < 3$?

Revenue is

$$R = QP = 36Q - 4Q^2 \text{ for } Q \leq 3$$

$$R = 30Q - 2Q^2 \text{ for } Q > 3$$



The example (NPD, Cont'd)

Marginal revenue is

$$\text{MR} = dR/dQ = 36 - 8Q$$

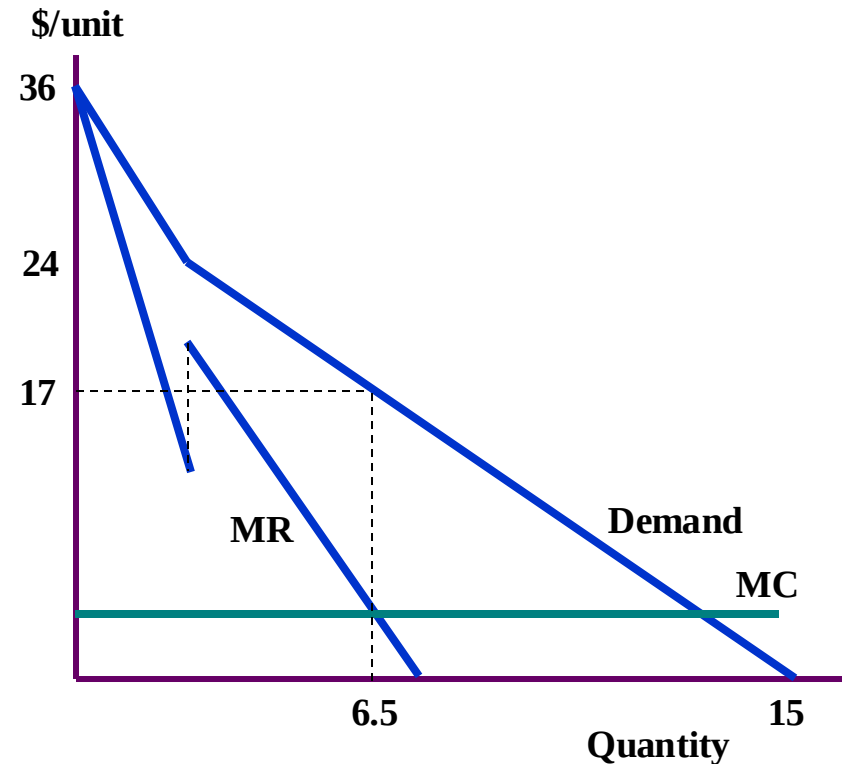
for $Q \leq 3$

$$\text{MR} = 30 - 4Q \text{ for } Q > 3$$

Set $\text{MR} = \text{MC}$

$$Q = 6.5$$

Price from the demand curve: $P = \$17$



The example (NPD, Cont'd)

Substitute price into the individual market demand curves:

$$Q_A = 9 - P/4 = 9 - 17/4 = 4.75 \text{ million}$$

$$Q_C = 6 - P/4 = 6 - 17/4 = 1.75 \text{ million}$$

$$\text{Aggregate profit} = (17 - 4) \times 6.5 = \$84.5 \text{ million}$$

The example: price discrimination

- **The firm can improve on this outcome**
- **Check that MR is not equal to MC in both markets**
 - $MR > MC$ in “children” market
 - $MR < MC$ in “adult” market
- **This requires that different prices be charged in the two markets**
- **How should quantity in the children market change to lower the MR in that market?**
- **Procedure:**
 - take each market separately
 - identify equilibrium quantity in each market by equating MR and MC
 - identify the price in each market from market demand

The example: price discrimination, Cont'd

Demand by adults:

$$P_A = 36 - 4Q_A$$

Marginal revenue:

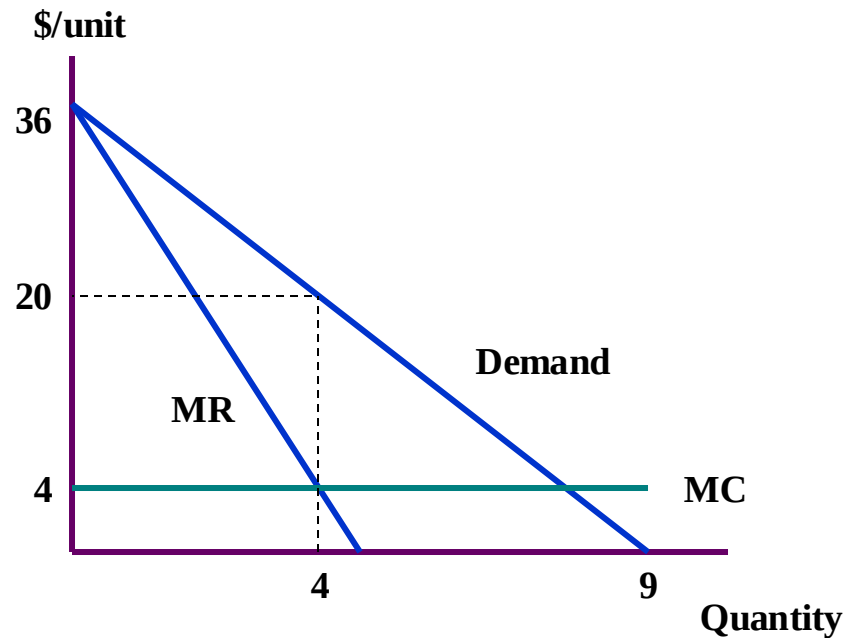
$$MR = 36 - 8Q_A$$

$$MC = 4$$

Equate MR and MC

$$Q_A = 4$$

Price from the demand curve $P_A = \$20$



The example: price discrimination, Cont'd

Demand by children:

$$P_C = 24 - 4Q_C$$

Marginal revenue:

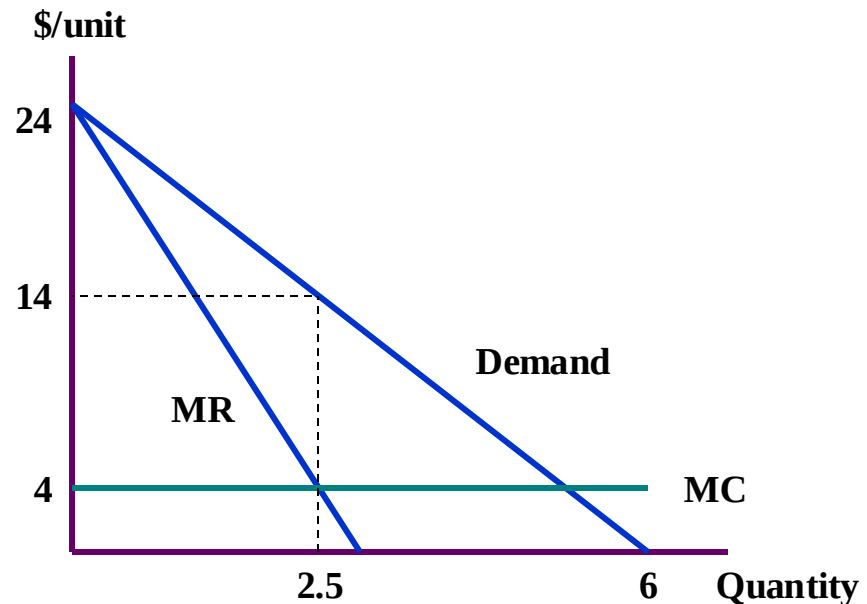
$$MR = 24 - 8Q_C$$

$$MC = 4$$

Equate MR and MC

$$Q_C = 2.5$$

Price from the demand curve $P_C = \$14$



The example: price discrimination, Cont'd

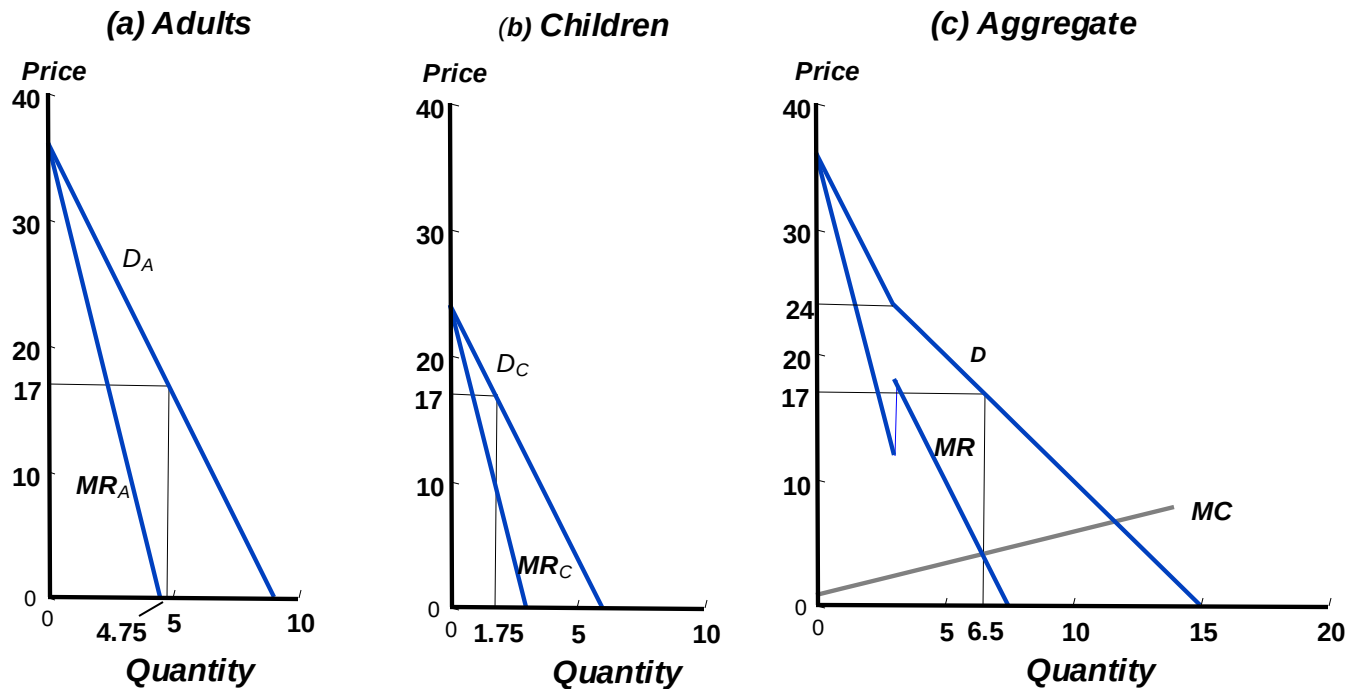
- **Aggregate sales are 6.5 million tickets**
 - the same as without price discrimination
- **Aggregate profit is**
 $(20 - 4) \times 4 + (14 - 4) \times 2.5 = \89 million
 - \$4.5 million greater than without price discrimination
 - that means 5.33% increase in profit

No price discrimination: non-constant cost

- **The example assumes constant marginal cost**
- **How is this affected if MC is non-constant?**
 - Suppose MC is increasing
- **No price discrimination procedure**
 - Calculate aggregate demand
 - Calculate the associated MR
 - Equate MR with MC to give aggregate output
 - Identify price from aggregate demand
 - Identify market demands from individual demand curves

The example again

Applying this procedure assuming that
 $MC = 0.75 + Q/2$ gives:

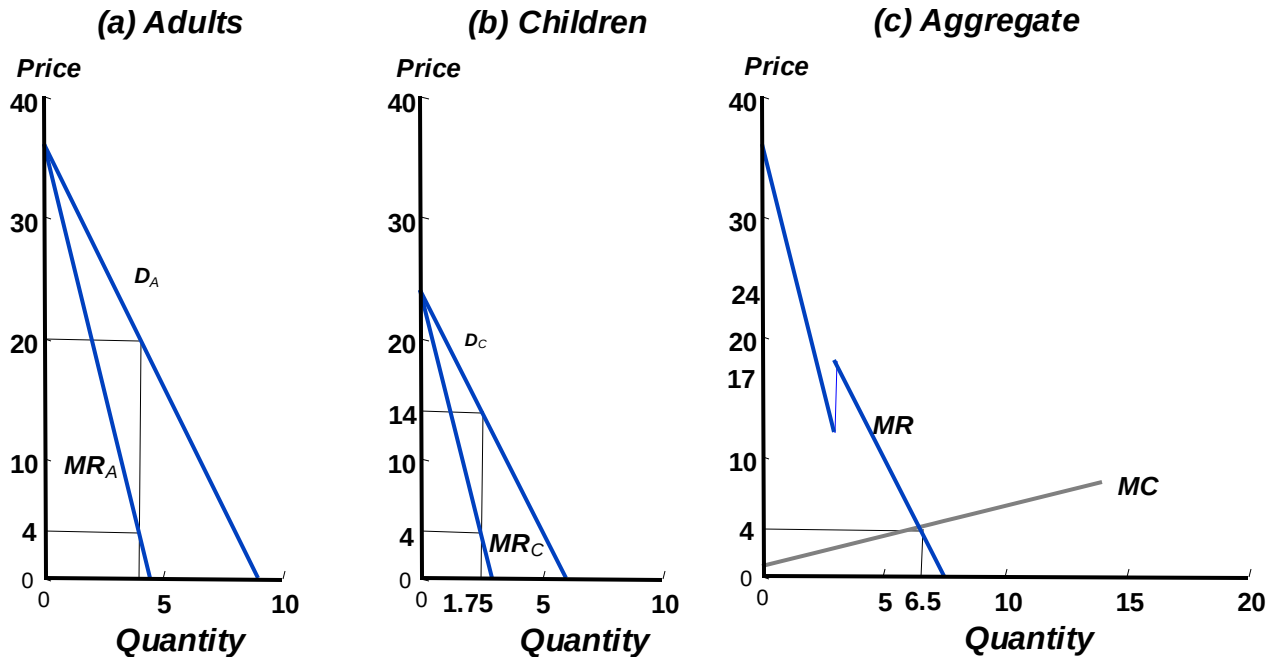


Price discrimination: non-constant cost, Cont'd

- **With price discrimination the procedure is**
 - **Identify marginal revenue in each market**
 - **Aggregate these marginal revenues to give aggregate marginal revenue**
 - **Equate this MR with MC to give aggregate output**
 - **Identify equilibrium MR from the aggregate MR curve**
 - **Equate this MR with MC in each market to give individual market quantities**
 - **Identify equilibrium prices from individual market demands**

The example again

Applying this procedure assuming that
 $MC = 0.75 + Q/2$ gives:



Some additional comments

- **Suppose that demands are linear**
 - price discrimination results in the same aggregate output as no price discrimination
 - price discrimination increases profit
- **For any demand specifications two rules apply**
 - marginal revenue must be equalized in each market
 - marginal revenue must equal aggregate marginal cost

Price discrimination and elasticity

Firm's profit:

$$\pi = \sum_{i=1}^N P_i d_i(P_i) - F - C \sum_{i=1}^N d_i(P_i)$$

where:

- *Number of markets: N*
- *Price in market i : P_i*
- *Quantity demanded in market i (depends on price): $d_i(P_i)$*
- *Marginal cost: C*
- *Fixed cost: F*

What could be wrong in the profit equation? So, what are the model risks here?

Price discrimination and elasticity, Cont'd

**Maximizing the profit with respect to price $\Rightarrow$
first order condition:**

$$\frac{d\pi}{dP_i} = d_i(P_i) + P_i d'_i(P_i) - C d'_i(P_i) = 0 \quad \text{for all } i = 1, \dots, N$$

$$\Rightarrow \frac{d_i(P_i)}{d'_i(P_i)} + P_i = C \quad (*)$$

Note that this is MR = MC:

$$\frac{d_i(P_i)}{\Delta d_i(P_i)/\Delta P_i} + P_i = C \Rightarrow \frac{\Delta P_i}{\Delta d_i} d_i + P_i = C \Rightarrow P_i' d_i + P_i = C,$$

which is the same as in slide 7.

Price discrimination and elasticity, Cont'd

Elasticity:

$$\varepsilon_i = -\frac{\Delta d_i}{d_i} \bigg/ \frac{\Delta P_i}{P_i} = -\frac{\Delta d_i(P_i)}{\Delta P_i} \frac{P_i}{d_i(P_i)} = -d'_i(P_i) \frac{P_i}{d_i(P_i)}$$

$$\Rightarrow -\frac{1}{\varepsilon_i} = \frac{d_i(P_i)}{d'_i(P_i)} \frac{1}{P_i} \quad \Rightarrow \quad 1 - \frac{1}{\varepsilon_i} = \frac{1}{P_i} \left[P_i + \frac{d_i(P_i)}{d'_i(P_i)} \right]$$

$$\Rightarrow P_i \left[1 - \frac{1}{\varepsilon_i} \right] = P_i + \frac{d_i(P_i)}{d'_i(P_i)} \quad \text{by substituting (*) from the previous slide}$$

$$\stackrel{(*)}{\Rightarrow} P_i \left[1 - \frac{1}{\varepsilon_i} \right] = C, \quad \text{for all } i = 1, \dots, N$$

Price discrimination and elasticity, Cont'd

Example with $N = 2$ (two markets): $P_1 \left[1 - \frac{1}{\varepsilon_1}\right] = P_2 \left[1 - \frac{1}{\varepsilon_2}\right]$, which gives:

$$\frac{P_1}{P_2} = \frac{1 - 1/\varepsilon_2}{1 - 1/\varepsilon_1}$$

So, the elasticities give the relative prices.

For instance, when ε_2 rises then P_1/P_2 rises as well.

Third-degree price discrimination 2

- **Often arises when firms sell *differentiated products***
 - female and male haircuts
 - academic license for cheaper services and products
 - business versus economy airfare (this can be viewed as a second-degree price discrimination as well)
- **Price discrimination exists in these cases when:**
 - “two varieties of a commodity are sold by the same seller to two buyers at different *net* prices, the net price being the price paid by the buyer corrected for the cost associated with the product differentiation.” (Phlips)
- **The seller needs an easily observable characteristic that signals willingness to pay**
- **The seller must be able to *prevent arbitrage***
 - e.g. require to show a documentation for student status

Product differentiation and price discrimination

- Suppose that demand in each submarket is $P_i = A_i - B_i Q_i$
- Assume that marginal cost in each submarket is $MC_i = c_i$
- Finally, suppose that consumers in submarket i do not purchase from submarket j
 - For instance: “I never buy paperbacks.”
- Equate marginal revenue with marginal cost in each submarket

$$A_i - 2B_i Q_i = c_i \Rightarrow Q_i = (A_i - c_i)/2B_i \Rightarrow P_i = (A_i + c_i)/2$$
$$\Rightarrow P_i - P_j = (A_i - A_j)/2 + (c_i - c_j)/2$$

It is highly unlikely that the difference in prices will equal the difference in marginal costs

Other mechanisms for price discrimination

- Impose restrictions on use to control arbitrage
 - Saturday night stay
 - no changes/alterations
 - personal use only (academic journals)
 - time of purchase (movies, restaurants)
- “Crimp” the product to make lower quality products
 - *Mathematica*®
- Discrimination by location

Discrimination by location

- Suppose demand in two distinct markets is identical:

$$P_i = A - BQ_i$$

- But suppose that there are different marginal costs in supplying the two markets:

$$c_j = c_i + t$$

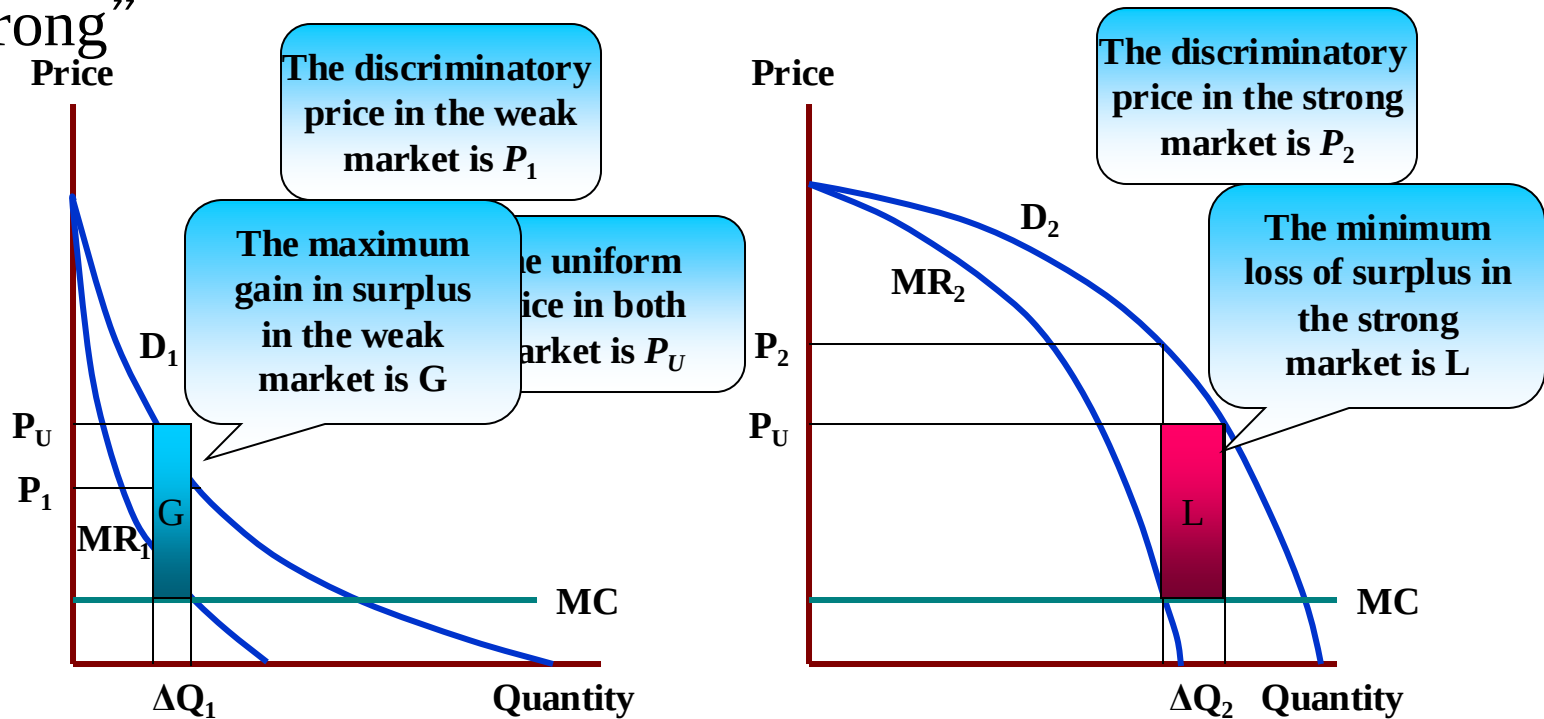
- For instance, food delivery
- Or amusement park operators that run 2 parks
 - Near identical demand (similar demographics)
 - Different operating cost (e.g. different labor costs)
- *Profit maximizing rule:*
 - equate MR with MC in each market as before
 - $\Rightarrow P_i = (A + c_i)/2; P_j = (A + c_i + t)/2$
 - $\Rightarrow P_j - P_i = t/2 \neq c_j - c_i$
- Difference in prices is not the same as the difference in costs even under identical demand functions!

Third-degree price discrimination and welfare

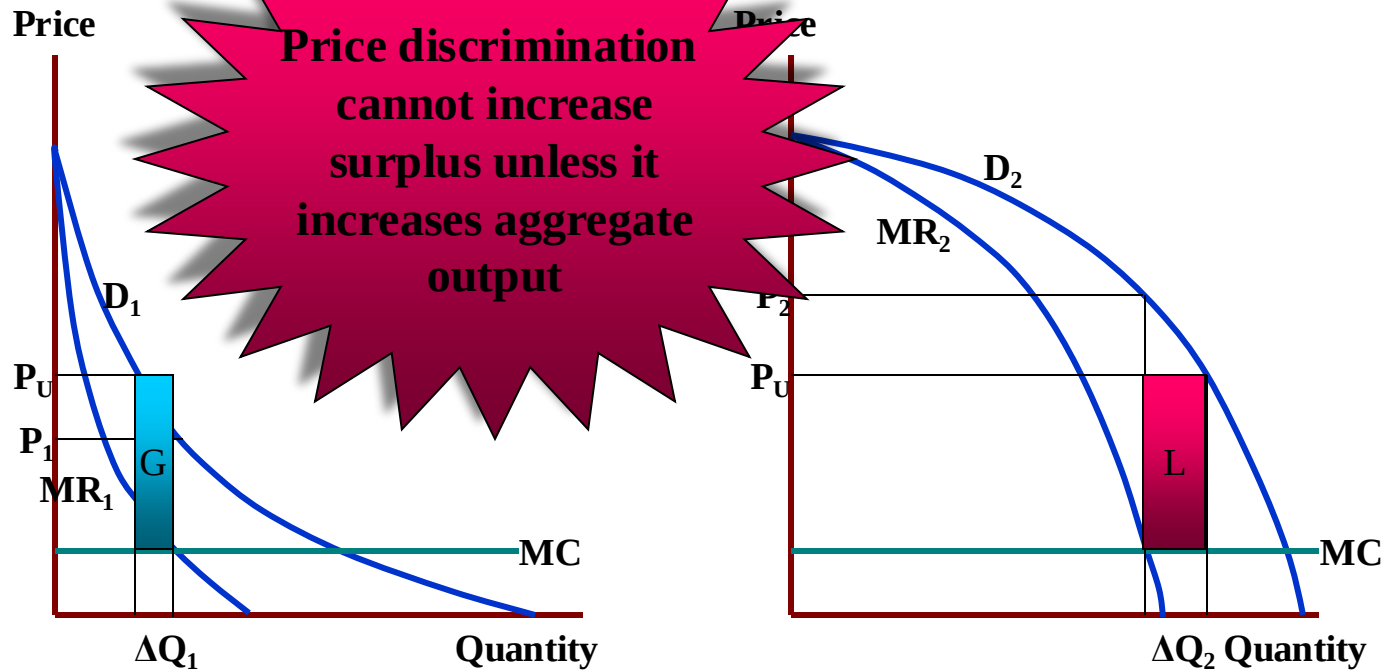
- **Does third-degree price discrimination reduce welfare?**
 - not the same as being “fair”
 - relates solely to efficiency
 - so consider impact on total surplus
- **Main point: *Price discrimination cannot increase surplus unless it increases aggregate output***

Price discrimination and welfare

Suppose that there are two markets: “weak” and “strong”



Price discrimination and welfare



It follows that $\Delta W \leq G - L = (P_U - MC)\Delta Q_1 + (P_U - MC)\Delta Q_2$

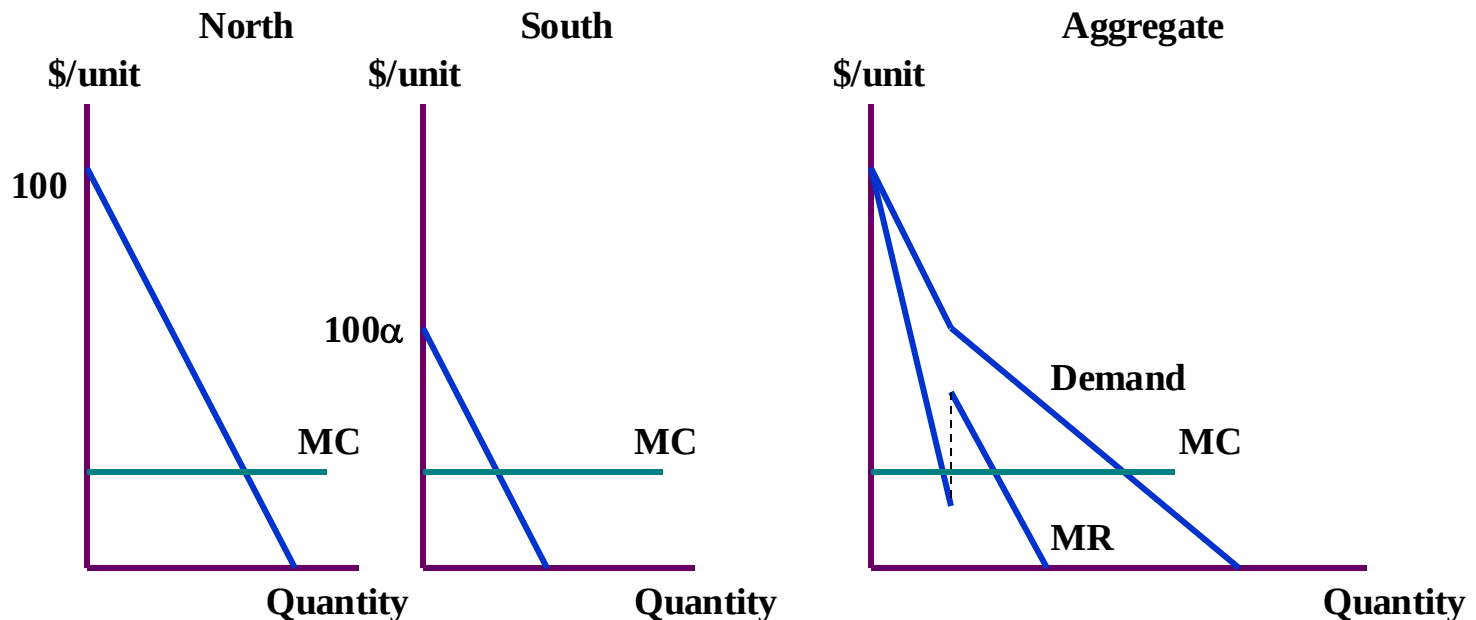
$$= (P_U - MC)(\Delta Q_1 + \Delta Q_2)$$

Price discrimination and welfare 2

- **Previous analysis assumes that the same markets are served with and without price discrimination**
- **This may not be true**
 - *uniform price is affected by demand in “weak” markets*
 - *firm may then prefer not to serve such markets without price discrimination*
 - *price discrimination may open up weak markets*
- **The result can be an increase in aggregate output and an increase in welfare**
- **Example: Vaccination prices are lower in developing countries**

New markets: example

- 2 areas – “North” and “South”
- “North” is a wealthy market (or country)
 - Higher Demand: $P_N = 100 - Q_N$
- “South” is a developing market (or country)
 - Lower demand: $P_S = 100\alpha - Q_S$
- Marginal cost to supply either market is \$20

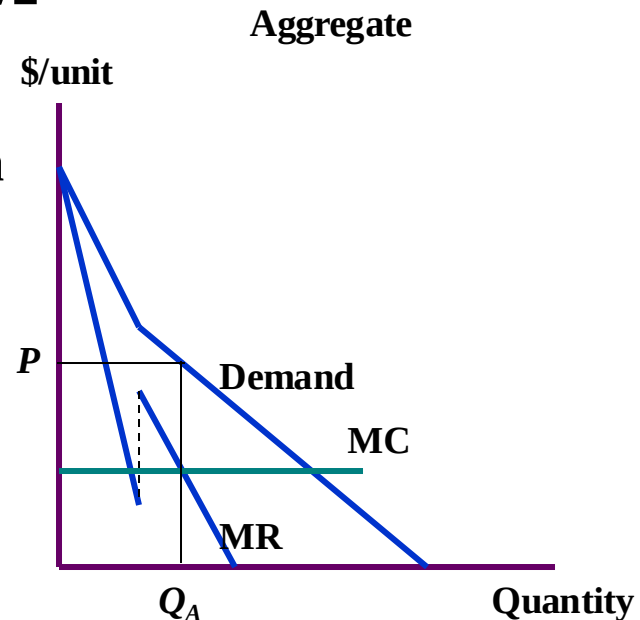


New Markets: the example 2

Aggregate demand is $P = (1 + \alpha)50 - Q/2$
provided that both markets are served

Equate MR and MC to get equilibrium
output $Q_A = (1 + \alpha)50 - 20$

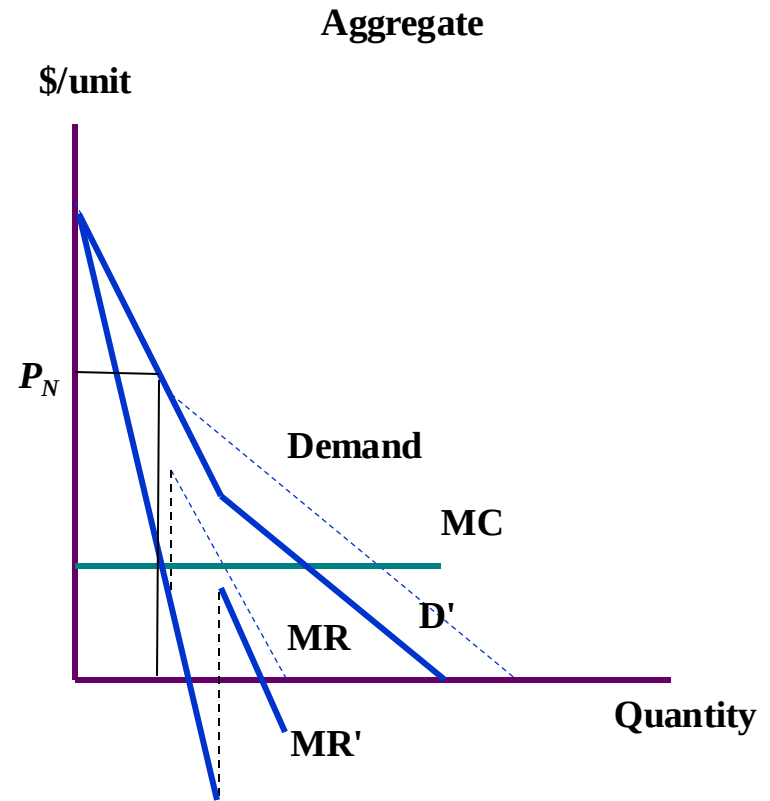
Get equilibrium price from
aggregate demand $P = 35 + 25\alpha$



New Markets: the example 3

Now consider the impact of a reduction in α :

- aggregate demand changes
- marginal revenue changes
- it is no longer the case that both markets are served
- the South market is dropped
- price in North is the monopoly price for that market



The example again

Previous illustration is too extreme:

- MC cuts MR at two points

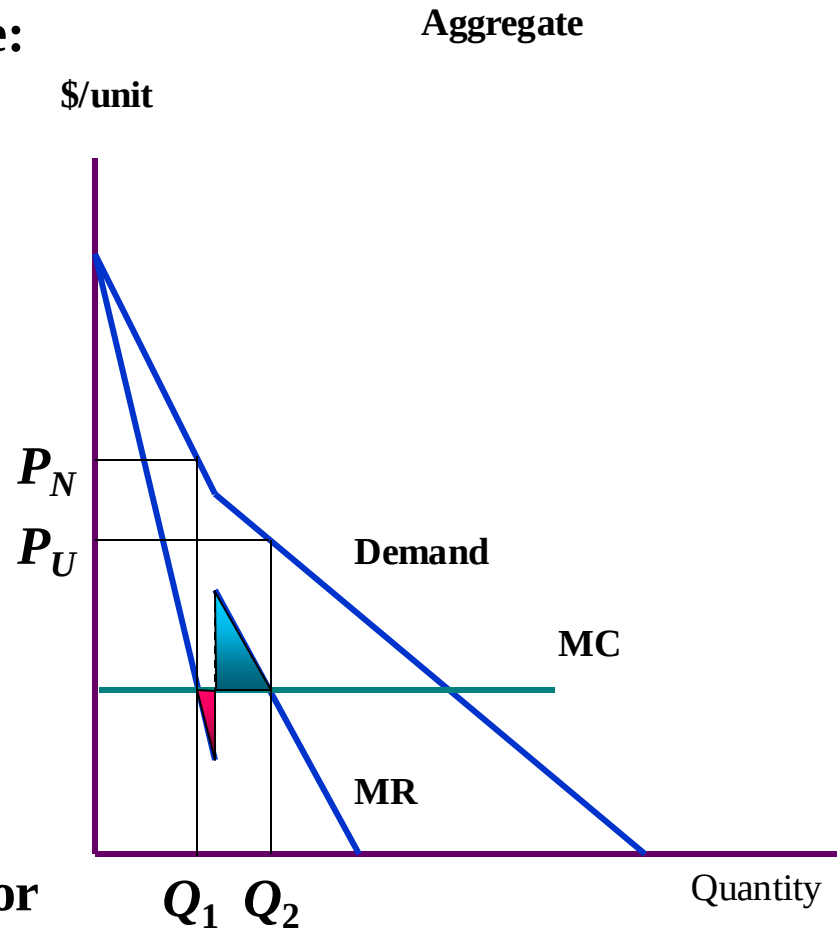
So, there are potentially two equilibria with uniform pricing:

- at Q_1 only North is served at the monopoly price in North
- at Q_2 both markets are served at the uniform price P_U

Switch from Q_1 to Q_2 :

- decreases profit by the red area
- increases profit by the blue area

If South demand is “low enough” or MC “high enough”, serve only North

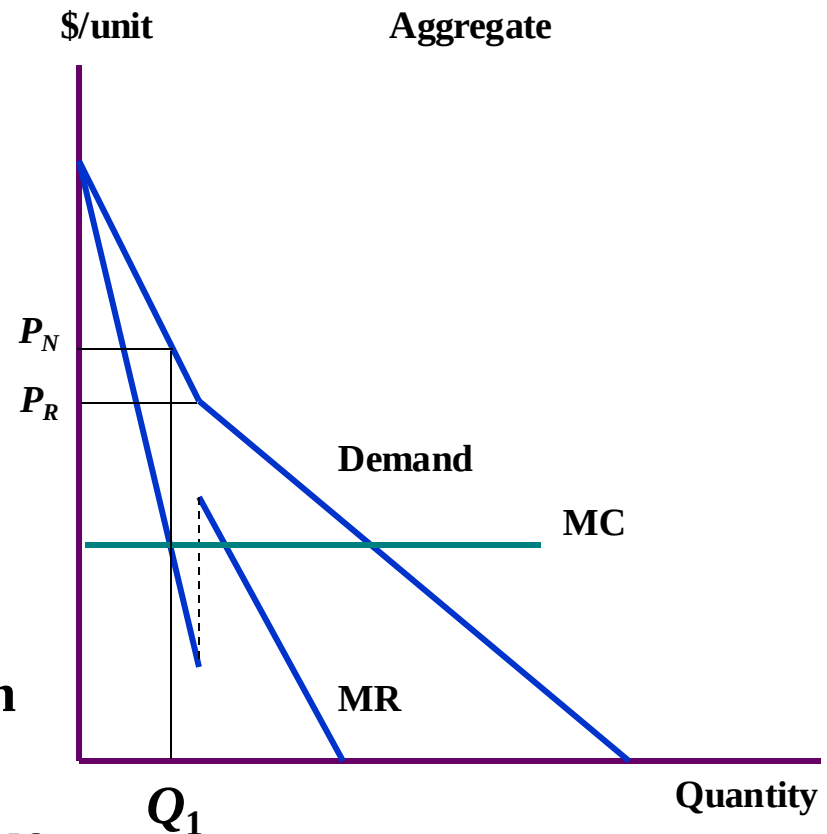


Price discrimination and welfare again

In this case only North is served with uniform pricing
But MC is less than the reservation price P_R in South
So, price discrimination will lead to South being supplied

Price discrimination leaves surplus unchanged in North
But price discrimination generates profit and consumer surplus in South

So, price discrimination increases welfare



Price discrimination and welfare one more time

- **Suppose only North is served with a uniform price**
- **Also assume that South will be served with price discrimination**
 - **Welfare in North is unaffected**
 - **Consumer surplus is created in South: opening of a new market**
 - **Profit is generated in South: otherwise the market is not opened**
- **As a result price discrimination increases welfare.**